Model Card — HealthGuard Health Claim Classifier

1. Model details

Field	Description
Model name	HealthGuard-v1
Task	Binary classification of health claims: True (verified) vs False (misinformation)
Model	TF-IDF (unigrams + bigrams, terms in fewer than 2 claims dropped) + Logistic Regression (max_iter=1000, balanced class weights)
Training data	PUBHEALTH train split, True and False labels only: 8,079 claims (5,078 True, 3,001 False). Mixture, unproven and 28 malformed rows excluded
Test data	PUBHEALTH test split, True and False labels only: 987 claims (599 True, 388 False)
Preprocessing	None beyond TF-IDF defaults. No stop-word removal and no negation handling
Libraries	Python 3 (Google Colab), scikit-learn 1.6.1, pandas, NumPy, matplotlib; Flask and joblib for the website server
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2. Intended use

- **Intended:** An educational demonstration of a classical NLP pipeline, and a transparent first check that flags English health claims for human review.
- **Out of scope:** Medical or treatment decisions; claims in other languages; social media posts; mixture or unproven claims; events after the dataset was collected. The model is not a fact-checker, and its output is not a medical verdict.

3. Overall performance

The model reaches 76.3% accuracy and 0.757 macro-F1 on the 987-claim test set, beating a majority-class baseline (60.7%) by 15.6 points.

Model	Accuracy	Precision	Recall	Macro-F1
Majority-class baseline (always True)	60.7%	—	100%	0.378
HealthGuard (TF-IDF + LR)	76.3%	0.837	0.756	0.757

Precision and recall are for the True (verified) class.

4. Performance across text subgroups

Key audit finding: accuracy drops by 12.8 points on long claims (more than 15 words) compared with short claims, and recall for verified long claims is only 19.8%, so most of them are labelled False.

Subgroup	How it was defined	Test size (n)	Accuracy	Macro-F1
Short claims	15 words or fewer	717	79.8%	0.754
Long claims	More than 15 words	270	67.0%	0.526

The model is least reliable on long claims, which is where most verified claims are missed. Accuracy alone hides this; macro-F1 and recall expose it.

5. Explainability check

Method 1 — Top features (Logistic Regression coefficients). Each word or phrase has a weight. Positive weights push a claim toward True; negative weights push it toward False. Topic words such as "coronavirus", "officials", "study" and "health" lean True, while function words such as "the", "is", "was" and "and" lean False, so the model partly learns writing style.

Top positive features (toward True)	Top negative features (toward False)
coronavirus ([insert weight])	the ([insert weight])
officials ([insert weight])	is ([insert weight])
study ([insert weight])	was ([insert weight])
health ([insert weight])	and ([insert weight])
mental health ([insert weight])	that ([insert weight])

Method 2 — [Insert second explainability method and its result, e.g. LIME on one misclassified claim, or the word contributions for one claim].

6. Limitations and ethical considerations

- **Main limitation:** The model does not understand word order or context. It relies on function words and misses many verified long claims (19.8% recall), so it should not be trusted on those without human review.
- **Data bias:** Claims come from fact-checking sites and include COVID-19 content, so the topic words the model learns reflect that period and those sources.
- **Label source:** Labels are fact-checker judgements, not clinical verification. Mixture and unproven claims were excluded, so the model cannot handle mixed or unverified claims.

- **Small audit sample:** The long-claim subgroup has 270 test claims, so its estimate is less precise than the short-claim one. The results also come from one train/test split with no cross-validation, and the 0.5 decision threshold is not tuned.

7. Recommendations

- Test removing common function words and compare the length audit again.
- Tune the decision threshold or add more long-claim training data, then re-run the same subgroup audit.
- Add cross-validation to check that the results are stable.
- Compare against a contextual model (for example, a fine-tuned transformer) using the same subgroup audit.
- Route low-confidence predictions to a human reviewer.